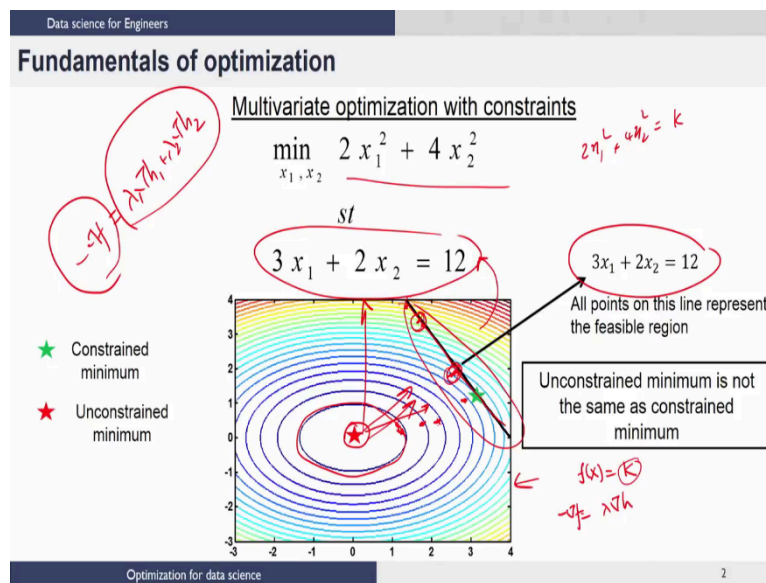


Data Science for Engineers
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Lecture – 27
Multivariate Optimization with Equality Constraints

Let us continue our lectures on optimization for data science. In this lecture we will look at multivariate optimization non-linear optimization. However, in the previous lectures we saw the unconstrained version of this problem. In this lecture we will look at problems of this type where we have what are called equality constraints. I will explain that is presently.

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And I am going to explain this using a very simple example, but before we dive into this example let us set some context and then ask the question as to why we should be interested in this in a course on data science. The reason why we should be interested in this in a course on data science; the reason why we are interested in constraints in optimization is as I mentioned before, we look at optimization from a data science viewpoint because we are trying to minimize error in many cases, when we try to solve data science problems. And when we minimize error, we said we could use some kind of gradient based

algorithm what we called as the learning algorithm to solve the problem.

In some cases while we are trying to optimize or minimize our error or the objective function, we might know some information about the problem that we want to incorporate in the solution. So, if for example, you are trying to uncover relationships between several variables and you do not know how many relationships are there, but you know for sure that certain relationships exist and you know what those relationships are, then when you try to solve the data science problem you would try to constrain your problem to be such that the known relationships are satisfied.

So, that could pose an optimization problem where you have constraints in particular equality constraints and there are several other cases where you might have to look at the constrained version of the problem while one solves data science problems. So, it is important to understand how these problems are solved. Once we look at equality constraint problems we will look at in-equality constraints which is even more relevant. For example, the algorithms for inequality can with inequality constraints are very useful in data science algorithm that is called support vector machines and so on. So we going to look at both equality and inequality constraints.

Now, let us look at this problem right here. So we are interested in mini-mizing the function here which is $2x_1^2 + 4x_2^2$ and like we discussed before this is an objective function in 2 variables x_1 and x_2 . So, real visualization of this would be in 3 dimensions where the objective function value is plotted in the z direction. However, we know that we can work with contour plots and if you look at this picture here, you see these contour plots. These are plots where each of this contour is a constant objective function contour and again you would see that these are ellipses because if you look at constant contour plots then you have $2x_1^2 + 4x_2^2$ equal k this you will see is an ellipse that is what is plotted here.

Now, if you look at the optimum value for this function and just by inspection you can see that the optimum value is 0 because this are functions where I have terms which are squares of the decision variables. So, there are 2 terms here $2x_1^2 + 4x_2^2$ and the lowest value that each one of these could take has to be 0 and that basically means the unconstrained minimum is at $x_1 = 0$ $x_2 = 0$ the objective value at that point is also 0.

So, which is what is shown here in this point right here. So, if I had no constraints I would say the optimum value is 0 and it is at 0 0 the star point here. Now let us see what happens if I introduce a constraint

in this case I am going to introduce a very simple linear constraint. So, let us assume that we have a constraint of this form which is $3x_1 + 2x_2 = 12$.

Now what this basically means is the following. You are looking for a solution to x_1 and x_2 which also satisfies this constraint that is what it means. So, though I know the very best value for this function is 0 from a minimization viewpoint, I cannot use that value because the 0 0 point might not satisfy this equation. So, you will notice that if I put $x_1 = 0$ $x_2 = 0$ it does not satisfy this equation. So, the unconstrained solution is not the same as the constrained solution.

Now, let us see how we solve this problem. To understand this we first start by representing this constraint in this 2 dimensional space. So, since this is a linear constraint you will see that it is a line which is here which is what is given by this constraint. Now since we said that whatever solution I get it should satisfy this constraint would translate to saying, I can only pick solutions from this line because only points on this line will satisfy this constraint. Now, you notice that you could pick many many points on this line. So, for example, you could pick this point and the objective function value at that point would be corresponding to a contour which intersects this point. So, if you pick a point here then you would have a corresponding objective function value. Now you could pick a point here then you would see that it would have another objective function value which would be corresponding to the constant function contour that intersects the line at that point.

So, you notice that as you pick different points on this line the objective function takes different values and what we are interested in is the following. Of all the points on this line, I want to pick that one point where the objective function value is the minimum. To understand this let us pick two points and see what happens. So, if I pick a point here and let us say I pick a point here and ask the question which one of these points is better from a minimization of the original function view point. The way to think about this is the following.

So, when I pick a point here I know that the objective function value would be based on the contour which intersects at that point and if you compare these 2 points you will see that this point is worse than this point, from a minimization viewpoint. This is because if you look at these contours these are contours of increasing objective function values and the contour that intersects this point is within the contour that intersects the line at this point. So, basically what that means, is because this is the direction of increasing objective function value the contour intersecting this point is inside the contour intersecting at this point. So, that basically means this function takes a lower value at this point on the line.

So, as you go along the line you see that the value keeps changing and my job is to find that particular point where the objective function value is the minimum. So, this is a basic idea of constrained optimization solution that we are looking for. The key point to notice here is that the unconstrained minimum is not the same as the constraint minimum. If it turns out that the unconstrained minimum itself is on the constraint line then both would be the same, but in this case we clearly see that the unconstrained minimum is different from the constrained minimum.

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The slide is titled "Fundamentals of optimization" and "Multivariate optimization with equality constraints". It contains the following content:

- At optimum (one equality constraint case)**

$$-\nabla f(\bar{x}) = \lambda^* \nabla h(\bar{x})$$
- In higher dimensions and when there are more than one equality constraint**

$$-\nabla f(\bar{x}) = \sum_{i=1}^I [\nabla h_i(\bar{x})] \lambda_i^*$$

Gradient lies in the space spanned by the normal of the gradients

Handwritten notes in red ink include:

- $f(x_1, x_2, \dots, x_n)$
- $h(x_1, x_2, \dots, x_n) = 0$
- $\min f(x_1, x_2, \dots, x_n)$
- $h(x_1, x_2, \dots, x_n) = 0$
- $-\nabla f = \lambda \nabla h$
- $\frac{\partial f}{\partial x_1} = \lambda \frac{\partial h}{\partial x_1}$
- $\frac{\partial f}{\partial x_2} = \lambda \frac{\partial h}{\partial x_2}$
- $\frac{\partial f}{\partial x_n} = \lambda \frac{\partial h}{\partial x_n}$
- $h(x_1, x_2, \dots, x_n) = 0$

So, when I have just only one constraint, how do I solve this problem? I will first give you the result and then explain the result again by going back to the previous slide and then showing you another viewpoint and then how that leads to this solution.

So, if you typically take a multivariate function $f(x)$ in the unconstrained version, let us assume that x is x_1 x_2 and x_n . We know that the optimum solution is $\text{grad } f = 0$ which basically means that $\partial f / \partial x_1$, $\partial f / \partial x_2$ all the way up to $\partial f / \partial x_n$ equal 0. So, this when I expand this further I will get equations $\partial f / \partial x_1$ equal 0 $\partial f / \partial x_2$ equal 0 $\partial f / \partial x_n$ equal 0 and so on.

Now, notice that if there are n variables there will be n equations of this form. So, I have n equations in n variables. So, I can solve for it and find a solution and once I find a solution to find out whether it is a maximum or minimum or a saddle point I calculate the second

derivative and then construct the hessian matrix and then depending on whether the hessian matrix is positive definite positive negative definite or semi definite and so on we can make judgments about whether the point is a minimum point maximum point and so on.

Now, let us see what happens if I am trying to solve a problem where I have to minimize $f(x)$ which is x_1, x_2 all the way up to x_n . But like I said before let us assume that I also introduced one constraint and I am going to introduce let us say a constraint which is of the form h of x_1, x_2 all the way up to $x_n = 0$. So, this is an equality constraint. So, we can always write a constraint in this form, even if you have some number on the right hand side you can always move to the left hand side and write this constraint as something equal to 0. So, basically my job is to find the minimum point for f subject to this constraint. I will just give you the result and then we will see how we get this result.

So, when I want to solve for this problem the result is in this case $\text{grad } f = 0$ itself gave us the result. In this case it will turn out that the result is the following. So, we can write the negative of ∇ has to be equal to some $\lambda \nabla$ of h . So, you can write this as negative or drop the negative and the sign of the value λ will take care of that, but I am writing in this particular form.

So, just to expand this basically says I have something like this I have domain is $\partial f / \partial x_1$ all the way up to $\partial f / \partial x_n = \lambda \partial h / \partial x_1$ all the way up to $\partial h / \partial x_n$. So, if we expand this further I am going to get n equations. In this case I got these equations to be 0 in this case I am going to get equations of the form $-\partial f / \partial x_1 = \lambda \partial h / \partial x_1$ will be one equation, the second equation will be $-\partial f / \partial x_2 = \lambda \partial h / \partial x_2$ and so on, the last equation will be $-\partial f / \partial x_n = \lambda \partial h / \partial x_n$.

So, now notice much like before I have n equations the difference being in the unconstrained case I had zeros on the right hand side in the constrained case I have these terms on the right hand side. Nonetheless these equations are in now $n + 1$ variable because I have my x_1, x_2 all the way up to x_n and I have also introduced a new variable λ right here. So, my equations are in $n + 1$ variables. I have only n equations at this point, but notice that I have one more equation that I need to use and that equation is the following if I find some solution x_1 to x_n which satisfies all of these equations.

Then that also has to satisfy the constraint. So, here we are only talking about the gradient form of the various functions, but the equation which represents the constraints also needs to be satisfied by

any solution that we get for the constrained optimization problem. So, with these n equations I will also get another equation which is that h of $x_1, x_2, \dots, x_n$ has to be $= 0$.

Now, you notice that in this case with one linear constraint I have $n + 1$ equation in $n + 1$ variables. So, I can solve this, so to reiterate the difference between the constrained and unconstrained case was, in the unconstrained case we had n equations in n variables, in the constraint case with just one constraint we have $n + 1$ equations in $n + 1$ variables.

Now, you might ask what happens, if there are there is more than one constraint there are let us say 2 constraints.

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Data science for Engineers

Fundamentals of optimization

Multivariate optimization with equality constraints

At optimum (one equality constraint case)

$$-\nabla f(\bar{x}^*) = \lambda' \nabla h(\bar{x}^*)$$

In higher dimensions and when there are more than one equality constraint

$$-\nabla f(\bar{x}^*) = \sum_{i=1}^l [\nabla h_i(\bar{x}^*)] \lambda_i^*$$

Gradient lies in the space spanned by the normal of the gradients

Optimization for data science

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So, we will see what happens if there are 2 constraints, so in this case we are going to minimize $f(x)$ and now let us say we have 2 constraints we are going to say $h_1(x) = 0$ $h_2(x) = 0$. In this case what is going to happen is the following the solution to the constrained optimization problem is going to be $-\nabla f = \lambda_1 \nabla f_{h_1} + \lambda_2 \nabla f_{h_2}$.

So, if you look at this and the one constraint case you will see the similarities. In the one constraint case we introduced one extra parameter, in the 2 constraints case the x introduced 2 extra parameters. In the one constraint case, we simply said $\text{grad } f - \nabla f$ equal $\lambda \nabla h$, in this case this $\lambda \nabla h$ becomes a sum of $\lambda \nabla h_1 + \lambda_1 \nabla h_1 + \lambda_2 \nabla h_2$ and so on.

Now, if there are 3 equality constraints, then you would have 3 terms here and if there are 1 equality constraints you will have

something like this here where this is sum of 1 terms. So, that part is clear the other part that we need to worry about is if I have enough equations to solve for all the variables when I have 2 constraints, that basically means I have introduced 2 new variables λ_1 and λ_2 . However, this equation irrespective of the number of equality constraints you have will always be n equations.

Because $\text{grad } f$ would be $\partial f / \partial x_1, \partial f / \partial x_2, \dots, \partial f / \partial x_n$ so these are all n by 1 vectors. So, this will just give me n equations. Nonetheless if I had 2 equality constraints I need to find the 2 extra equations which are directly given by the constraints. So, since the optimum point has to satisfy both the constraints, I get one extra equation $x_1 - x = 0$ and the other extra equation is $2x = 0$. So, if you have 2 equality constraints in n variables I will have $n + 2$ variables and $n + 2$ equations and we will always find this to be true because if I had 3 equality constraints, I will have $n + 3$ variables which would be x_1 to $x_n, \lambda_1, \lambda_2, \lambda_3$ and this gradient equation will always give me n equations and the 3 extra constraints would have given me the 3 extra equations.

So, I will have $n + 3$ equations and $n + 3$ variables which can be directly generalized to this form where I have 1 equality constraints. So, let us see in the single constraint case how we get an expression like this that that would be easy to see in the single constraint case. This expression where we have the sum of these terms is slightly more complicated to understand. I am not going to do that. I am going to give you an intuitive feel for why this is true in the single equation case and once you understand that with a little bit more effort you should be able to think about why it should be true for more equality constraints and so on.

So, we go back to the previous slide and when I was discussing this slide and explaining how equality constraints affect the optimum solution. I said there are many points on this line which could all be feasible solutions. Feasible solutions meaning those are all solutions which can satisfy this equation. Nonetheless, there are some points out of those or one point which would give me the lowest objective function value. So, when we looked at candidate solutions for this optimization problem we were looking at the points on this line that that we are interested in because that is a constraint.

Now, let us take a slightly different viewpoints and then look at the same problem from an objective function viewpoint. Now from an objective function viewpoint if you did not constrain me at all and you said you could do anything you want, then I would pick this point as a solution. Now when you look at this point and then say well this is a

best point I have let me find out whether it satisfies my constraint and then you substitute this point into this and then you figure out it does not satisfy the constraint.

So, you say look I have to do something because I am forced to satisfy the constraint. So, you will say let me lose a little bit in terms of an objective function perspective and then see whether I can meet the constraint.

So, when we say I want to lose a little bit basically you know as we mentioned before these are contours where the objective function value increases and those are actually not good from a minimization viewpoint. So, while I am here since the constraint is not satisfied, I am willing to lose a little to see whether I can satisfy the constraint and maybe I go to a point here and then this is a constant objective function contour point. So, if I am willing to give up something that basically means I am going and sitting on different points on this contours and as I am pushed away and away from this minimum point I am losing more and more in terms of the objective function value. That is I am increasing the objective function value. Now logically if you keep extending this argument you will see that, let us say this is the first point I moved here which is basically worse than this because you see this is a contour which is going to be outside of this contour, but I moved here I made my objective function worse, but I still am not satisfying the constraint. So, I give some more I come to this point and I see a contour and this point is worse than this because this contour is outside this contour and if I extract this argument let us say I keep making things worse and the only reason I am making these worse is because I am forced to satisfy this equality constraint.

So, I come up to let us say here and this is still a contour which is much worse than my original solution, but it is still not enough for me to satisfy my constraint. So, if you keep repeating this process, you are going to find a contour here where I touch this line for the first time. So, when I touch this line for the first time is the point at which for the first time, when I give up my objective functions value, I am also able to satisfy the constraint. Now once I find a contour like that which touches this line then there is no incentive for me to go further beyond because going further beyond would mean I would be making my objective function worse.

So, when I just touched that line that is the best compromise because that is where I become feasible for the first time and going any further would only make my objective function worse. So, geometrically what this would mean is I keep making my objective function worse till a contour just touches this line. So, at that point this line will become a tangent to that contour and remember what that contour is that contour is $f(x) = k$ for some k

So, we have to choose a k in such a way that this contour, when I have the constraint, the constraint becomes a tangent to this contour and optimum point. This geometric fact when you represent it as equations, you would get the form that I showed which is - grad of f is λ times grad of h . So, this is for the one constraint case, now when you have many equality constraints while it is not as easy to see that as it is in the single equality constraint case the statement that - grad f let us say if I have 2 is $\lambda_1 \text{ grad } h_1 + \lambda_2 \text{ grad } h_2$ this statement basically says that this gradient negative of the gradient has to be written as a linear combination of grad of h_1 and grad of h_2 and that has a geometric interpretation.

So, that is the reason why we get these conditions for optimization with equality constraints the key point that I want you to remember is that as you have more and more equality constraints you will have to introduce more and more parameters $\lambda_1 \lambda_2$ and so on. However, there will always be enough equations and variables.

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Data science for Engineers

Fundamentals of optimization

Multivariate optimization

$$\min_{x_1, x_2} 2x_1^2 + 4x_2^2 \quad \checkmark$$

st

$$3x_1 + 2x_2 - 12 = 0 \quad \checkmark$$

First order condition

$$\begin{pmatrix} -4x_1 \\ -8x_2 \\ (3x_1 + 2x_2 - 12) \end{pmatrix} = \begin{pmatrix} \lambda \\ \lambda \\ 0 \end{pmatrix}$$

solving $\rightarrow$

$$\begin{bmatrix} x_1^* \\ x_2^* \\ \lambda^* \end{bmatrix} = \begin{bmatrix} 3.27 \\ 1.09 \\ -4.36 \end{bmatrix}$$

$\frac{\partial f}{\partial x_1} = 4x_1 = 3\lambda$
 $\frac{\partial f}{\partial x_2} = 8x_2 = 2\lambda$

$-\nabla f = \lambda \nabla h$
 $h(x) = 0$
 $-\nabla f \cdot \begin{pmatrix} \frac{\partial h}{\partial x_1} \\ \frac{\partial h}{\partial x_2} \end{pmatrix} = \begin{pmatrix} -4x_1 \\ -8x_2 \end{pmatrix} = \begin{pmatrix} 3\lambda \\ 2\lambda \end{pmatrix}$

$\begin{pmatrix} \frac{\partial h}{\partial x_1} \\ \frac{\partial h}{\partial x_2} \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$

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So, let us look at a numerical example to bring all the ideas together. We are going to use the same example that we had looked at in the previous slides. So, we are trying to minimize $2x_1^2 + 4x_2^2$ as objective function subject to this constraint. So, we wrote the following equations for the optimum point for identifying the optimum point we said - ∇ of f has to be $\lambda \nabla$ of h and then we have h of $x = 0$. So, this is what we wrote let us do the computation so that we understand this.

So to calculate ∇f which is basically $\partial f / \partial x_1$ $\partial f / \partial x_2$. So, if you look at this objective function. So, $\partial f / \partial x_1$ will be simply $4x_1$ and $\partial f / \partial x_2$ will be $8x_2$. So, we have ∇f so negative grad of f would be negative. Let us look at grad of h . So, h is this equation so if I do $\partial h / \partial x_1$ $\partial h / \partial x_2$. So, this is going to be ∇h $\partial h / \partial x_1$ will be simply 3 and $\partial h / \partial x_2$ will be simply 2 so we have this.

So, now, let us see what this equation becomes. So, this equation you would see is if you put this in a bracket and you will see this easily. So, you have the first equation is $-\partial f / \partial x_1 = \lambda \partial h / \partial x_1$ and we have $-4x_1 = 3\lambda$, similarly the second equation would turn out to be $-8x_2 = 2\lambda$ and this equation is basically the same equation as the constraint equation that we have here. And as we mentioned before we thought the constraint that would have been 2 variables and you would have got 2 equations which would have been $\nabla f = 0$.

But with an equality constraint we have added a new parameter λ . So, we need 3 equations in x_1 x_2 and λ we do have these 3 equations here and when you solve these 3 equations you will get this solution and this is your optimum solution in the constrained case which is different from the optimum solution in the unconstrained case which would have been 00. So, in other words we have given up on the value of the objective function.

However, this is a point which would satisfy this equation of the line. So, that is how we deal with equality constraints in an optimization problems. I already described how these are useful or why we should study them in the first place from a data science perspective. With this I will conclude this lecture and in the next lecture we will look at how we handle inequality constraints and I will also explain why we are interested in understanding how optimization problems are solved with inequality constraints from a data science perspective.

Thank you and I look forward to seeing you again in the next lecture.

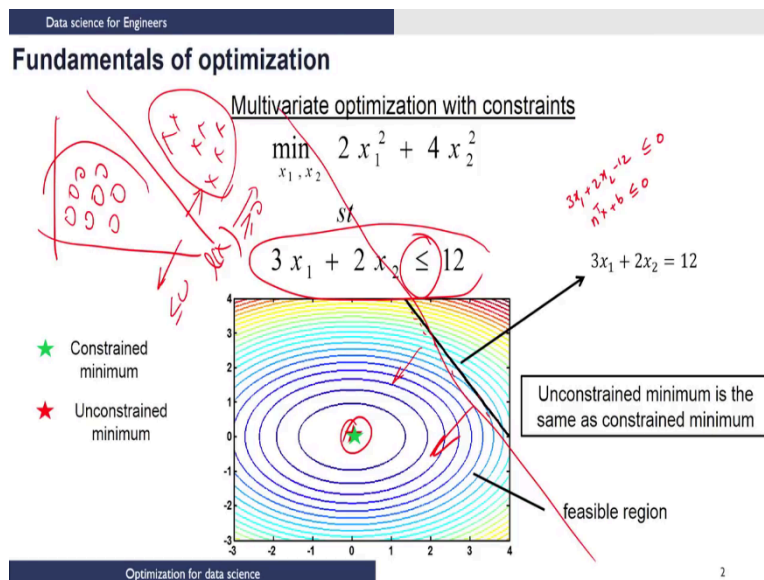
Data Science for Engineers
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Lecture – 28
Multivariate Optimization with Inequality Constraints

We come to the last topic in this series of lectures on optimization for data science. Till now we saw how to solve unconstrained multivariate optimization problems. Then we saw how to solve multivariate optimization problems with equality constraints. Now we move on to multivariate optimization problems with inequality constraints. Before I start explaining some of the key ideas in these types of problems, let us look at why we might need this optimization technique in data science.

Remember in one of the earlier lectures I talked about data for people who like south Indian restaurants and so on.

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So, if you remember that example I said there are a group of people who might like certain type of restaurant there might be a group of people who do not like that kind of restaurant and so on. Then if we

were to do a classifier which probably is a line like this then, as we are trying to solve an optimization problem to identify a classifier like this, we have to impose the constraint that all these data points will have to be on one side of the line and all of these data points will have to be on the other side of line and from our lecture on half spaces and hyper planes in linear algebra we know that if this equation is something like this which is linear equation, then it might be that if the normal is in this direction then this direction is said that if I substitute a value of this point into this it is greater than equal to 0 and on this side it is less than equal to 0 with 0 being the line.

So, now, notice that for each point if we were to write the condition in terms of the equation of the line and then you would see that these become inequality constraints. So, there may be as many in equality constraints there are points and so on.

So, you see why we might be interested in imposing inequality constraints and optimization problems from a data science viewpoint. A more sophisticated version of this idea is what is used in one of the data science algorithms called support vector machine. Though we will not study that technique in this first course on data science for engineers, I just wanted to point out that this class of optimization problems is very important from a data science viewpoint.

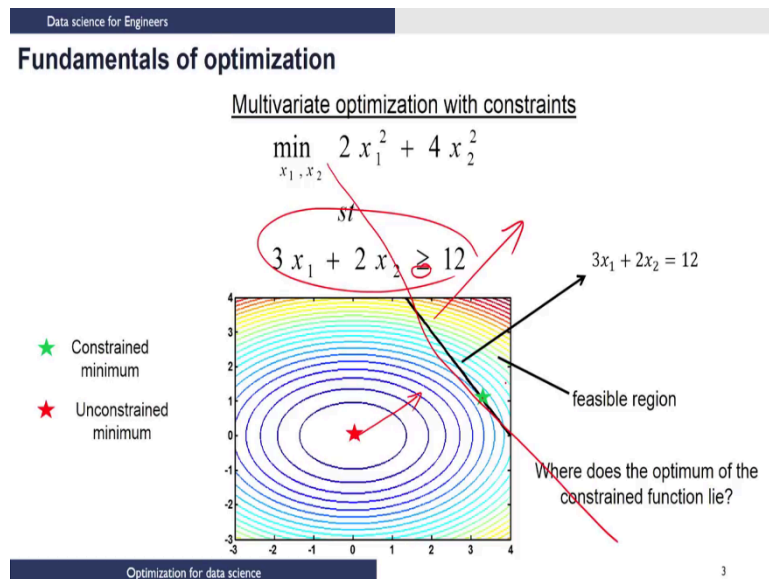
Now, let us go back to the same example that we had before, where we had the equality constraint and then we tried to solve the optimization problem and then just make that equality into an inequality. So, in this case let us assume what was equal to 12 in the last lecture has now become less than equal to 12 and let us understand intuitively what happens to problems this of this type. Now in the previous case we said when we have an equality constraint we said we are interested in any point on this line as a candidate solution these are all called the feasible points and of all of these points we were trying to pick the point which will give me the minimum objective function value.

Now, when I have the optimization problem where I have this inequality and let us assume this is less than equal to in this picture what we have plotted is that the original unconstrained optimum or the unconstrained minimum is still the red star. Now if you look at this and then say this less than equal to 12 it can be basically rewritten as $3x_1 + 2x_2 - 12 \leq 0$ and remember that this is of the form in transpose $x + b \leq 0$. So, it is going to be one half space depending on how n is defined. In this case it will turn out that this is the half space that is represented by this equation.

So, the difference between the equality constraint and the inequality constraint is the following, in the equality constraint we had every point on this line being a feasible solution when you make this as a inequality constraint. Then what happens is if you think of this line is extended all the way, any point to this half space now becomes a feasible solution and because every point in this half space is a feasible solution the unconstrained minimum also becomes a feasible point so the constrained minimum and unconstrained minimum are the same so this is an interesting thing to see.

So we saw that in the case of equality the unconstrained and constrained minimum were different, but when we made this into an inequality with the less than equal to sign we see that the constrained and unconstrained minimum are the same points.

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Now, let us try and see what happens if I flip the sign and then said this is greater than equal to 12. Then what would happen is if you were to extend this line all the way, then the feasible region is to this side. Now, you asked the question where does the optimum lie? You will notice that again we know the best solution is here and as we move away from this solution, we will see that we are losing out on the objective function value and as before we know any point on this line or to the side is a feasible point and any point on this line is also feasible because of this equality sign.

Now, making the same arguments that we made in the case of the equality constraint problem, we will see that I give up on my optimality, that is I keep going through contours of larger and larger

size where the optimum value keeps increasing and when a contour particular contour touches this line exactly at one point then I have a feasible point which is going to satisfy this constraint, the equality part of the constraint. So, it is satisfying the general constraint and that is the worst I have lost in terms of how much my objective function has increased its value by.

Anything more would be unnecessary because if I move little further I am going to make my objective function value worse. However, there is no need to do that because I already found a feasible point here. So, this case the constrained minimum becomes the same as the minimum that we achieved with the equality constraint case. So, you see that depending on what type of inequality these different things can happen.

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Data science for Engineers

General formulation

Multivariate optimization

$$\min_x f(\bar{x})$$

st

$$h_i(\bar{x}) = 0, i = 1, \dots, m$$

$$g_j(\bar{x}) \leq 0, j = 1, 2, \dots, l$$

Necessary condition for $\bar{x}^*$ to be the minimizer

KKT conditions has to be satisfied

Sufficient condition

$\nabla^2 L(\bar{x}^*)$ has to be positive definite

Optimization for data science

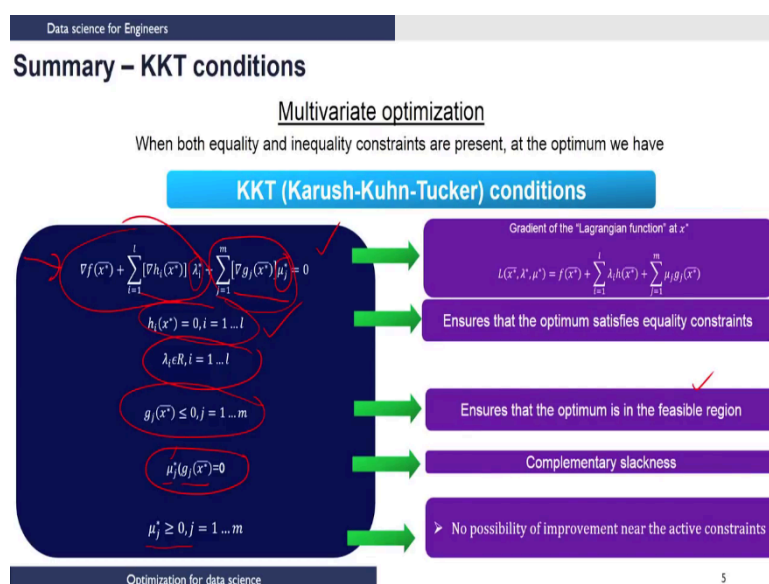
Now, from a general mathematical formulation viewpoint, we are going to generalize this. What I am going to show is I am going to show you the general conditions. Now for this course we might not be using this later in any data science algorithm. Nonetheless this forms a basic idea for more sophisticated data science algorithms like svm and so on. So, it is worthwhile to pay attention and then understand this completely. So, general multivariate optimization problem we can say has both equality and inequality constraints.

So, I have the objective function, I have m equality constraints and l inequality constraints. Notice that we can always write the equality condition in this form where I have 0 on the right hand side because if you have something on the right hand side I simply move it to the left hand side and get a 0 on the right hand side. Similarly, any condition

inequality condition whether it is greater than equal to 0 or less than equal to 0 I can always put it in this form. If the original condition is already in less than equal to form then it is the same as this if it is in the greater than equal to form then what I do is I multiply by a negative and then I have this condition has less than equal to 0.

So, now, I call this my constraint so I can again put it in this form. Now this becomes a little more complicated formulation and I am going to show you the conditions for this in the next slide. Which will look a little complicated in terms of all the math that is there. What I am going to do is I am just going to simply read out the conditions in the next slide and then we will take a particular example and then demonstrate how the conditions work.

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These are the conditions for multivariate optimization problem with both equality and inequality constraints to be at it is optimum value and let us look at this carefully. Remember when there was only equality constraints, we had this part of the expression where we had ∇ of f + linear sum of λ times ∇h for each one of the equality constraints.

Now, when you also have inequality constraints, what happens to this equation is, just like how we added a single parameter for every equality constraint, we add a parameter for each inequality constraints. So, if there are m inequality constraints there will be m parameters and in typical optimization books and literature people use λ as a scalar multiple for each one of these constraints. So, if there are l equality constraints, there will be l lambdas and people use the nomenclature of

μ for the inequality constraints. So, if we have m μ s there will be, there will be m μ s corresponding to the m inequality constraints.

So, the difference between this condition in the equality and inequality case is that for every one of these inequality constraints you add more linear combinations of μ times δg . So, look at this, this is the same form of as this except that I used λ here and μ here, but I take a ∇ of h and ∇ of g . That is the first set of conditions then much like how we had the constraints equality constraints also as part of conditions in the previous case, I am going to have the optimum solution satisfy all of this equality constraints.

Now the λ is some real number, so as many real numbers as there are equality constraints and much like how I still need to have this equality condition satisfied the optimum point, I need to have the inequality constraint also to be satisfied by the optimum point. So this ensures that the optimum point is in the feasible region. Now this real differences between the equality constraint condition and the inequality constrained situation shows up. We also have additional constraints, which are of this form, these are called complementary slackness condition.

So, what this says is if you take a product of the inequality constraint and the corresponding μ_i then that has to be 0. Basically what it means is either μ_j is 0 in which case this is free to be any value such that this condition is satisfied or this is 0 in which case I have to compute a μ and the μ that I compute has to be such that it is a positive number or it is greater than equal to 0. So, this condition is there to ensure that whatever optimum point that you have, there is no possibility of improvement -any more improvement- from the optimum point so that is the reason why this condition is there.

Now just keep in mind that if you are seeing course on optimization for the first time it is not very easy or natural to understand this constraints right away. However, what we are going to do is in the next slide we will take an example and then show you how these things work. One thing that I want you to keep in mind is if we had let us say an unconstrained optimization problem objective function in n variables, I always look at whether the optimum conditions have enough equations and variables for me to be able to solve the system of equations and clearly you know in the unconstrained case you have n equations and n variables and I clearly made the point in the equality constraint case that for every equality constraint you add an extra parameter.

However, you will have enough equations and variables because when you write the first condition which is of this form you will get the n equations and you will get as many equality constraints that need to

be satisfied as there are lambdas. So that also works out properly. Now we will see whether the same thing happens in the case of inequality constraints.

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Data science for Engineers

Summary – KKT conditions

Multivariate optimization

- In general it is difficult to use the KKT conditions to solve for the optimum of an inequality constrained problem (than for a problem with equality constraints only) because we do not know a priori which constraints are active at the optimum.
- Makes this a combinatorial problem ✓
- KKT conditions are used to verify that a point we have reached is a candidate optimal solution.
- Given a point, it is easy to check which constraints are binding.



Optimization for data science

There is a specific problem in solving this -what are called the KKT conditions-the conditions that I showed in the previous slide are called the KKT conditions. It is not easy to solve the KKT conditions directly in the inequality case because of the complimentary slackness condition which says either μ could be 0 or z could be 0. So, we have to make a choice as to which is 0 so that makes this a combinatorial problem.

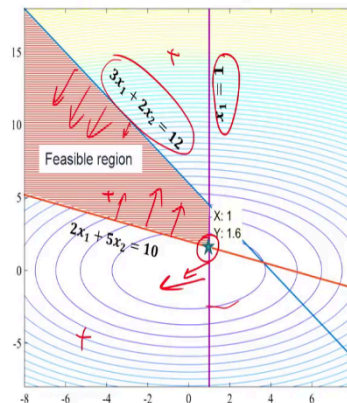
So, in general the KKT conditions are generally used to verify if a solution that we have is an optimal solution. However, optimization algorithms will have different ways of solving this problem and there are methods called penalty methods, active set methods and so on, we are not going to talk about those methods here. Nonetheless I just want you to understand how the solution works out in an analytical case.

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Fundamentals of optimization

Multivariate optimization-quadratic programming

$$\begin{aligned} \min_{x_1, x_2} \quad & 2x_1^2 + 4x_2^2 \\ \text{s.t.} \quad & 3x_1 + 2x_2 \leq 12 \\ & 2x_1 + 5x_2 \geq 10 \\ & x_1 \leq 1 \end{aligned}$$



Optimization for data science

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Let us take a look at a numerical example to bring together all the ideas that we have described till now. In this particular case it is a multivariate optimization problem which is actually called quadratic programming and this is called quadratic programming because the objective function is quadratic and the constraints are linear. Those types of problems are called quadratic programming problems.

I think this is the same objective that we have been using till now in the several examples. So, let us say this is the objective function and let us assume that we have constraints of the form shown here. Let us assume the first constraint is $3x_1 + 2x_2 \leq 12$, the second constraint is $2x_1 + 5x_2 \geq 10$ and the third constraint is $x_1 \leq 1$. Now I am not doing anything more to this problem, but nonetheless I just want you to remember that to be consistent with whatever we have been saying this should be converted to a less than equal to constraint which we will see how that happens in the next slide.

Now let us look at this pictorially. Remember that the value of the objective function is going to be plotted in the z direction coming out of the plane of the screen that you are seeing. So, the representation of that are these contours that we see here, which are constant objective function contours. So, I am just trying to see and explain how this picture speaks to both the objective function and the constraints. So, the objective function is actually represented in this picture as this constant value contours. So, if I am moving on this the objective function value the same we have repeated this several times.

Now in this $x_1 x_2$ plane, let us look at how these constraints look. So, I have this equation which is the equation of a line. So, when we

talk about the first constraint which is less than equal to 12, then any point to this side of the line is a feasible point. Now, when we look at this constraint here then the equation of the line is $2x_1 + 5x_2 = 10$ and whenever I have something greater than $= 10$, this region is a feasible region and this is the $x_1 = 1$ line and $x_1 \leq 1$ would be this region.

So, if you put all of these regions together the only region which is feasible is shaded in brown colour here. So, if you take a point any point in this region it will be satisfying this constraint because it is to this side of this line, it will satisfy the second constraint because it is to the side of the line and it will satisfy the third constraint because it is to this side of the line. Notice that if you take any point anywhere else you will not be feasible. For example, a point here would violate constraint 1, but it would be feasible from constraint 2 and 3 viewpoint nonetheless all the constraints have to be satisfied. Now similarly if you take a point here while it satisfies constraint 1 and 3 it will violate constraint 2.

Now, if you notice the optimum point is going to lie here and we are going to try and find out this value through the conditions that we described in the last few slides.

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Data science for Engineers

Fundamentals of optimization

Multivariate optimization-quadratic programming

$$\begin{aligned} \min_{x_1, x_2} \quad & 2x_1^2 + 4x_2^2 \\ \text{s.t.} \quad & 3x_1 + 2x_2 \leq 12 \Rightarrow (a) \\ & 2x_1 + 5x_2 \geq 10 \Rightarrow (b) \\ & x_1 \leq 1 \Rightarrow (c) \end{aligned}$$

- Lagrangian
$$L(x_1, x_2, \mu_1, \mu_2, \mu_3) = 2x_1^2 + 4x_2^2 + \mu_1(3x_1 + 2x_2 - 12) + \mu_2(10 - 2x_1 - 5x_2) + \mu_3(x_1 - 1)$$
- First order KKT conditions

$$\begin{aligned} 4x_1 + 3\mu_1 - 2\mu_2 + \mu_3 &= 0 \\ 8x_2 + 2\mu_1 - 5\mu_2 &= 0 \\ \mu_1(3x_1 + 2x_2 - 12) &= 0 \\ \mu_2(10 - 2x_1 - 5x_2) &= 0 \\ \mu_3(x_1 - 1) &= 0 \\ \mu_i &\geq 0 \end{aligned}$$

Handwritten notes: $10 - 2x_1 - 5x_2 \leq 0$, $3x_1 + 2x_2 - 12 \leq 0$, $x_1 \leq 1$, $3x_1 + 2x_2 - 12 = 0$, $10 - 2x_1 - 5x_2 = 0$, $x_1 = 1$.

Optimization for data science

So, what we do here is the following. So, we do something called a Lagrangian, which basically will boil down to the kind of conditions that we have been talking about I will explain this quickly. So, we take the objective function value or the objective function expression here and then there are no equality constraints here, there are only inequality constraints. I said we had one parameter age for each of these inequality constraints.

So, this is already in a form that is palatable to us which is less than equal to form. So, I just simply write this as μ_1 times $3x_1 + 2x_2 - 12 \leq 0$. So, this is already in the less than = form, when I look at the second equation, I want to make it into a less than = form then what I do is I said $10 - 2x_1 - 5x_2 \leq 0$. So, if I put this here this is into the less than = form and then corresponding to this if I have μ_1 corresponding to this μ_2 and corresponding to this I have μ_3 I have $x_1 - 1 \leq 0$ so you see that term here.

Then what you do is, you differentiate this with respect to x_1 and x_2 . You will get the first 2 conditions that we have been talking about for the con-strained case with equality constraint, unconstrained case and so on. So, the 2 equations for the 2 decision variables so, when you differentiate this whole expression with respect to x_1 $4x_1$ comes out of this term right here and this is only a function of x_2 . So, differential with respect to x_1 will become 0 and I will have $3\mu_1 x_1$ when I differentiate this with respect to x_1 I get $3\mu_1$ and from the second term I will get $-2\mu_2$ and from the third term I will get μ_3 . So, $3\mu_1 - 2\mu_2 + \mu_3 = 0$. So basically this equation you have and then when I differentiate the same expression with respect to x_2 .

So, I get $8x_2$ from this here and from the second term I get $2\mu_1$, I get $-5\mu_2$ from this term and from this term I get nothing because it is only a function of x_1 . So, I get this equal to 0 so, this basically gives you the condition that that we have talked about which is of the form in the equality constraint case remember I said you have $-\nabla$ has to be $= \sigma \lambda$ I ∇ of h_i and in the inequality case you also add $+\sigma \mu_i \nabla$ of g_i if g_i are the inequality constraints.

So, you kind of back out these 2 equations from this constraint and you have 2 equations here because ∇ of x is of size 2 by 1 h is 2 by 1 2 by 1 because there are 2 decision variables and these are scalars and this is a linear weighted sum. So, if you notice all of this, you see that I have 2 equations. However, I have 5 variables that I need to compute, I need to compute a value for x_1 , I need to compute a value for x_2 and then I need to compute μ_1 , μ_2 and μ_3 . So, let us see how we do that. We go back and add the complementary slackness conditions. Also keep in mind that other than this we also have to make sure that whatever solution we get still has to satisfy the 2 inequality constraints also.

So, whatever solution we get should still be such that $2x_1 + 2x_2 - 12$ is less than $= 0$ and $10 - 2x_1 - 5x_2$ is less than $= 0$. So, these 2 also have to be valid, but because these are inequality constraints we cannot actually use them to solve for anything. Once we solve for these variables we have to still verify whether this conditions are satisfy or not. So, going back to the complementary slackness condition we saw that we will have μ times g is 0. So, this is corresponding to the first

inequality constraint, this is corresponding to the second inequality constraint and this is corresponding to the third inequality constraints and we also have to have this μ_i greater than $= 0$. So, all of these conditions have to be satisfied for an optimum point.

Now, it is interesting to notice something here for let us just take this as an example already we have 2 equations, I have already mentioned that the 2 equations are here. So, I am looking for another 3 equations to compute μ_1 , μ_2 , and μ_3 . So, that will be a total of 5 equations and 5 variables. So, for this to be 0 you could say in this equation either μ_1 is 0 or whatever is inside the bracket is 0. So, we could say one possibility is whatever is inside the bracket is not 0 in which case μ_1 has to be 0. So, the key point that I want to make here is if we say whatever is in the bracket is not 0, then that automatically gives me the value for μ_1 . So, out of the 5 variables here I have already computed one, similarly if I say whatever is inside here is not 0 then μ_2 has to be 0 and whatever is inside here is not 0 μ_3 has to be 0 then I have already computed values for μ_1 , μ_2 , μ_3 .

Then I could substitute those values back into this equation and I have 2 equations I can calculate x_1 and x_2 . So, this is one option. So, in one of the previous slides I had mentioned that this becomes a combinatorial problem because we could also assume that this goes to 0. So, this term goes to 0 and now let us assume actually this is nonzero, this is nonzero, let us see what happens to that case do we have enough equations and variables.

So, in this case we will have 1 equation, 2 equations, the third equation will be μ_2 has to be 0 because we have assumed this is not 0 the fourth equation has to be μ_3 is 0 because we have assumed this is not 0. Now the fifth equation becomes the one which we have assumed which is the term inside the bracket is 0. So, in which case again I have 1 2 3 and then $\mu_2 = 0$ $\mu_3 = 0$ as 5 equations in 5 variables. You could for example, assume that this and this are 0 in which case I have to compute μ_2 and μ_1 and then let us say this is not 0 then μ_3 is 0, but in that case also you will have 1 equation, 2 equation then 0 is 1 equation and then 0 is 1 equation. So, again I have 5 equations and 5 variables

So, whichever assumption you make as far as these equations are concerned you will have enough equations and enough variables. However, for some of those choices when you actually find a solution and try to plug this back into this right here it might not satisfy this. So, in which case that is not a viable option for us from an optimization viewpoint. And in some cases this might be satisfied, but the μ that you calculate out of the equations you get might not be positive. So, it might get negative μ in which case again this is not an optimum solution. So, let us see how this happens for this example.

Now, I am going to use one notation here so, that we can understand the table in the next line. So, whenever I assume that an equality constraint is exactly satisfied; that means, when I say $3x_1 + 2x_2 - 12 = 0$, then we say this constraint is active it is active because the point is already on the constraint. If I take a point here then that is not on the line so, that is basically less than $= 0$ so, I will say it is inactive. So, for every constraint I can say whether it is active or inactive. So, if this constraint is active; that means, $3x_1 + 2x_2 - 12$ is 0. If this constraint is active; that means, $2x_1 + 5x_2 - 10 = 0$ and if this constraint is active; that means, $x_1 = 1$.

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Data science for Engineers

Fundamentals of optimization

Multivariate optimization-quadratic programming

Sl.no	Active (A) /inactive (I) constraints (a) (b) (c)	Solution (x, μ)	Possible optima (Y/N)	Remark
1	A A A	Infeasible	N	Equations do not have a valid solution.
2	A A I	$x = [3.6364 \ 0.5455]$ $\mu = [-5.2 \ -1.45 \ 0]$	N	$x_1 \leq 1$ is not satisfied, $\mu_1 < 0$, $\mu_2 < 0$
3	A I A	$x = [1 \ 4.5]$ $\mu = [-18 \ 0 \ 50]$	N	$\mu_1 < 0$
4	I A A	$x = [1 \ 1.6]$ $\mu = [0 \ 2.56 \ 1.12]$	Y	All constraints and KKT conditions satisfied
5	A I I	$x = [3.27 \ 1.09]$ $\mu = [-4.36 \ 0 \ 0]$	N	$x_1 \leq 1$ is not satisfied
6	I A I	$x = [1.21 \ 1.51]$ $\mu = [0 \ 2.45 \ 0]$	N	$x_1 \leq 1$ is not satisfied
7	I I A	$x = [1 \ 0]$ $\mu = [0 \ 0 \ -4]$	N	$2x_1 + 5x_2 \geq 10$ is not satisfied
8	I I I	$x = [0 \ 0]$ $\mu = [0 \ 0 \ 0]$	N	$2x_1 + 5x_2 \geq 10$ is not satisfied

Optimization for data science

So, in the example that we are considering right now, there are 3 inequality constraints and as I mentioned in the previous slide if the inequality constraint is exactly satisfied that does it becomes an equality constraint then we call that an active constraint and if the constraint is not exactly satisfied we call it as an inactive constraint. And now we have 3 inequality constraints and each of these constraints could be either active or inactive.

So, there are 2 possibilities for each of these constraints and since we have 3 constraints there are 2 to the power 3 possibilities which = the 8 possibilities that we have here. So, what I am going to do in this case is we are going to enumerate all possibilities for you to get a good understanding of how this approach works when you have inequality constraints. So, let me pick let us say a couple of rows from this table to explain the ideas behind how this works and what we are going to do is in the next slide we are actually going to see graphically what each of this case means.

So, let us look at the first row for example, here the choice we have made is all the 3 inequality constraints are active. That means, they all become equality constraints. Notice something interesting remember there are 2 decision variables. So, each inequality constraint is basically representing one half space for a line and when they become active each of these constraints become an equality constraint each of them become line.

Now when all 3 are active then we have 3 equations in 2 dimensions right. So, there are only decision variables x_1 and x_2 , but I have 3 equations in those 2 variables and from our linear algebra lectures we know that when we have more equations than variables in many cases we are not going to find a solution for the 2 variables which will satisfy all the 3 equations. So, in this case you cannot solve this problem it is infeasible because though I have enough equations a subset of these equations 3 equations are in 2 variables and I cannot find a valid solution. We will understand what this is geometrically in the next slide.

Let us look at some other condition here. Let us pick for example, row 5. So, if you look at row 5, we have made a choice that the first constraint is active, the second constraint is inactive and the third constraint is inactive, that basically means the first constraint equal to 0, the second and third constraints have to be less than equal to 0, which needs to be tested after we go through the solution process. Now much like how I described before in this case also we will be able to find 5 equations and 5 variables and we can solve for x_1 and x_2 which is shown here and we have solved for μ which is shown here.

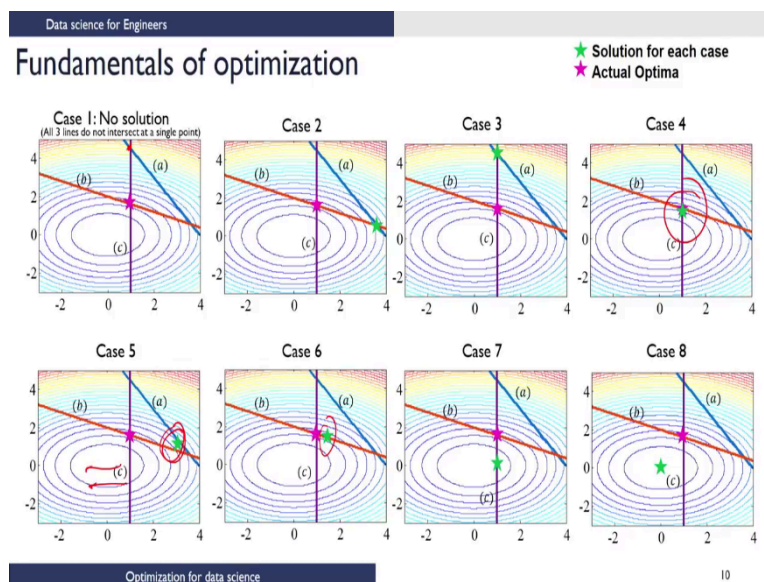
Now if you just look at this solution, you will you will say it seems to satisfy everything because I have a solution for x_1 and x_2 and I have μ_1, μ_2, μ_3 where μ_1 is - 4.36 and μ_2 and μ_3 are 0. However, when you look at this μ you will see that one of the μ s is negative. Which basically means that this cannot be an optimum point based on the conditions we showed in a couple of slides back, not only that on top of it when you actually put these 2 values into the constraint x_1 is less than equal to 1 it is not satisfied because x_1 is 3.27.

So, if you take this row for example, it looks like both this μ being positive is not satisfied and this constraint is also not satisfied. An interesting thing to note here is we have to go back and check only the constraints that we have assumed to be inactive because the active constraint is already at 0. So, whatever solution again will automatically satisfy the active constraint. Let us say row 6 for example, here we have made a choice that the first constraint is inactive, the second constraint is active and the third constraint is inactive.

Again we will get 5 equations and 5 variables and we will look at the solution here. It again looks good from the x view point. We have solved for x_1 and x_2 , 1.21 and 1.51. Unlike the previous case in this case the μ s also looked good because μ s have to be positive. So, I have 0 to 0.450. So, it looks like this, this is a good candidate for an optimum point. But once you have satisfied all of this you have to still go back and look at whether the inactive constraints are satisfied by this solution point right here and the inactive constraint here is x_1 has to be less than equal to 1, but if you notice that 1.21 does not satisfy this constraint. So, this cannot be an optimum either.

When you look at row 4, where we have assumed constraint 1 is inactive and 2 and 3 are active, I get a solution x_1 1.6. I get all μ s to be positive which is also one of the conditions for an optimum point and then I have to only verify this constraint here because other 2 are active constraints and they are providing equations for us to solve so they are like they are going to be valid. Now when you put these 2 values into the first constraint you will check that it actually satisfies the inequality also. So, this is a case where all constraints and KKT conditions are satisfied. So, this is the optimum point so, the optimum solution is 1.6 which is what we had indicated in the slide with the geometry of this problem.

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So, let us look at each of this case in terms of where the optimum lies and how we interpret this. So, if you take the case one where we took all the constraints to be active we said we have 2 variables and 3

equations and that is not satisfiable in general. Geometrically what this means is we want the point to lie on all the 3 lines at the same time because we have assumed all 3 lines are active concerned. That means, all of them have to be equal to 0.

So, you find that you cannot get a point where the all 3 lines equations are satisfied. So, if I want to satisfy 2, I will get a point here nonetheless it would not satisfy the other constraint and so, on. So, this is the case of not all 3 lines intersect so we do not have a solution here. Similarly you can look at each of these cases and you can see what happens in each of these and you will notice that only in case 4 will you have the solution that you got from the Kuhn Tucker condition and the actual optimum being satisfied.

In every other case you will see that there is some problem or other. So, if you go back I think we looked at case 5 and 6 then the solution for case 5 is this, the solution for case 6 is this and we said these two are not good solutions because they violate the condition x_1 is less than equal to 1 which basically seen here because the feasible region is to this side of line x_1 is less than equal 1, but this point is violating this constraint and here again you see that this point is violating this constraint.

So, in summary what you need to know is that in the unconstrained case it is very clear that I just simply write this ∇ of f is 0 as the condition and I will have enough equations and variables there will be n equations and n variables. In the constraint case with equality constraints I will get the same n equations, but the form will be slightly different we write that as $-\nabla f$ is $\sigma \lambda_i \nabla$ of h_i and since we add as many variables as there are equality constraints and since the equality constraints have to be satisfied those give you the extra equation. So, you will have enough equations and variables.

In the inequality constraint case, the first n equations come from a very similar form where $-\nabla f$ is $\sigma \lambda_i \nabla$ of $h_i + \sigma \mu_i \nabla$ of g_i that gives you n equations and corresponding to every one of those λ corresponding to equality constraints you will get so many equations which are the equalities have to be satisfied. The only complication comes in when you have these inequality constraints where either you know you have can have one or the other be 0 that is what we call as complementary slackness.

So, we have this μ times g going to be 0. So, if g is 0 we call that as an active constraint in which case we have to calculate the μ corresponding to it and in the optimum point μ will be greater than equal to 0. Now depending on the form in which you write whether you write all of these constraints or as inequality constraints less than equal to or greater than equal to and also in terms of whether you write

the original equation as $-\nabla f$ equal to this sum on the right hand side or ∇f equal to sum on the right hand side the sign of μ in different textbooks and different papers might be reported as either they have to be positive or they have to be negative.

So, you have to be careful about the conditions when you look at those, but if you stick to this type of writing the equations, where you write $-\nabla f = \sum \lambda_i \nabla h_i + \sum \mu_j \nabla g_j$ and if you write all the constraints as less than equal to inequality constraints, then μ s have to be positive for the point to be an optimum point which is what we saw here.

As I mentioned before while this unconstrained case is a very very important case in machine learning algorithms such as svm and so on the constrained case I mean we might not be using this quite a bit in this course. Nonetheless for the sake of completeness and for the sake of giving the foundations for understanding other data science algorithms that outside of this course that you might go and study.

I have also described the key ideas behind how to solve constrained optimization problems when you have equality and inequality constraints. Keep in mind that while I have shown you the conditions, I have not shown a proof or a derivation of these conditions in a formal manner. The equality constraint case I appealed to intuition to tell you why the conditions turn out to be the way they are. However, all of this can be formally proved and you can derive these conditions based on formal mathematical arguments.

So, with this we conclude this portion on optimization for data science now we have all the tools that we need to understand data science problems. The next set of lectures what I am going to do is I am going to introduce to you different types of data science problems that we encounter, how do we think about these data science problems. Is there some formal way of thinking about these problems; that we can use to solve a variety of problems and then we will move on to regression as a function approximation tool and we will look at different clustering techniques that are used in data science and then we will finally conclude with one particular technique called principle component analysis which is very useful for engineers and end with more general example of how one solves real life data science problems.

So, I hope to see you continue these lectures and understand more of data science.

Thank you.